

Personae: Automated Customer Segmentation with LLM-Generated Personas

Machine Learning (Summer 2026) — Final Project Proposal

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Problem Description

Businesses accumulate large volumes of customer data but struggle to translate it into a usable understanding of *who their customers actually are*. Marketing and product teams typically rely on analysts to manually define customer segments — a slow, subjective process that often produces groups based on intuition rather than behavior. As a result, smaller organizations frequently operate with no segmentation at all, sending the same message to every customer.

We propose **Personae**, a web application that automatically discovers natural customer groups from a raw behavioral dataset and describes each group as an actionable marketing persona. The specific question we aim to answer is: *given an unlabeled table of customer behavior, can an unsupervised pipeline recover distinct, interpretable segments, and can a large language model translate each segment's statistical profile into a clear, grounded persona that a non-technical marketer could act on?* The product takes a customer dataset as input and returns labeled segments, a visualization, and a one-paragraph persona per segment (e.g., "*The Cash-Advance Revolver — carries a high balance, relies heavily on cash advances, and rarely makes purchases*").

Data Summary

We use the **Credit Card Dataset for Clustering** [1], a public dataset of **8,950 anonymized credit-card customers** described by **17 numeric behavioral features** spanning roughly six months of activity. Features include account balance, total and one-off purchases, installment purchases, cash-advance usage, purchase and cash-advance frequencies, credit limit, payments, and the fraction of balance paid in full. The data contains real-world imperfections — `MINIMUM_PAYMENTS` is missing for 313 customers and `CREDIT_LIMIT` for one — providing a genuine preprocessing task. Because the features are anonymized financial behavior rather than demographics, the resulting personas are grounded in *how customers actually use their accounts*, which we consider a strength: segments reflect verifiable behavior rather than assumed lifestyle. The dataset is sufficiently large and feature-rich to support meaningful clustering while remaining tractable for an interactive application.

Methods

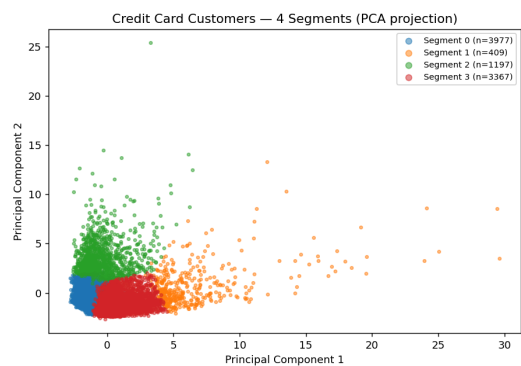
Our pipeline combines three techniques from the course. First, **data preprocessing**: we impute the missing values and standardize all 17 features so that no single high-magnitude field (e.g., balance) dominates distance computations. Second, **Principal Component Analysis (PCA)** [2] reduces the standardized features for visualization and to mitigate redundancy among correlated fields. Third, **K-means clustering** [3] partitions customers into segments, with the number of segments selected objectively using the elbow method and the **silhouette coefficient** [4]. Finally, each segment's mean feature profile is passed to a large language model, which generates a named, plain-language persona constrained to reference only the segment's actual statistics. The application will let a user upload a customer file and receive segments, an interactive PCA scatter plot, and the generated personas. Anticipated challenges include selecting the number of segments in a principled way, keeping LLM-generated personas factually grounded in the cluster statistics rather than fabricated, and generalizing the pipeline to datasets with differing column schemas.

Preliminary Results

We have completed initial setup and exploration. After imputing missing values and standardizing the features, PCA indicates that 10 components capture 90% of the variance, with the first two components (used for plotting) capturing 47.6%. A K-means sweep over $k = 2-8$, evaluated by silhouette score, supports a small number of well-separated groups; at $k = 4$ the model yields four clearly interpretable segments: a large low-activity group (3,977 customers), a high-spending group averaging \$7,682 in purchases (409), a cash-advance-reliant group averaging \$4,522 in cash advances with low purchases (1,197), and an active everyday-spender group (3,367). These segments are visually distinct in PCA space, confirming that the dataset supports the proposed approach.

References

- [1] A. Bhasin. *Credit Card Dataset for Clustering*. Kaggle, 2019. <https://www.kaggle.com/datasets/arjunbhasin2013/ccdata>
- [2] I. T. Jolliffe. *Principal Component Analysis*, 2nd ed. Springer, 2002.
- [3] J. MacQueen. "Some methods for classification and analysis of multivariate observations." *Proc. Fifth Berkeley Symposium on Mathematical Statistics and Probability*, 1967.
- [4] P. J. Rousseeuw. "Silhouettes: A graphical aid to the interpretation and validation of cluster analysis." *Journal of Computational and Applied Mathematics*, 20:53–65, 1987.



Appendix, Figure 1. Four customer segments projected onto the first two principal components.